

Emotion Analysis Using BRIGHTER + EthioEmo Datasets

Evaluating Data Augmentation Techniques for Multilabel Emotion Classification in Low-Resource African Languages

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1 Introduction

Emotion analysis (EA) automatically identifies textual emotions, but African languages remain underrepresented due to linguistic diversity and data scarcity. Consequently, models struggle with multilabel classification, where a single text conveys multiple overlapping emotions. To mitigate this, this project investigates data augmentation, specifically back-translation and paraphrasing, to artificially expand training datasets. By evaluating these techniques alongside qualitative error analysis, this research seeks to improve multilabel emotion classification for low-resource African languages and build more inclusive, responsible NLP systems.

2 Background

Emotion analysis (EA) identifies specific, often overlapping emotional states (e.g., anger, joy) rather than basic positive or negative feelings, requiring multilabel classification for real-world accuracy (Plaza-del-Arco et al., 2024). However, the vast annotated data required by modern EA models creates a barrier for African languages, which remain heavily low-resource (Adebara and Abdul-Mageed, 2022). While recent Afrocentric efforts have advanced foundational NLP, complex tasks like emotion classification remain under-explored and rarely optimised for typological nuances like complex morphology and code-switching (Adebara, 2024).

To address data scarcity, researchers use data augmentation, specifically back-translation and paraphrasing, to artificially expand training sets. These approaches excel in low-resource environments by increasing lexical diversity and maintaining semantic fidelity, often outperforming AI models without prior examples (Radlinski et al., 2025). However, their application to multilabel emotion classification in African languages remains unexplored. Despite benchmarks like the EthioEmo

dataset, the field is still vastly under-resourced. This project bridges this gap by evaluating back-translation and paraphrasing on multilabel emotion tasks for underrepresented African languages, providing essential insights to build inclusive NLP models.

3 Research Questions

1. How can data augmentation strategies (such as back-translation and paraphrasing) improve multilabel emotion classification for African languages?
2. How can existing NLP methods be adapted or applied effectively in low-resource African language contexts for emotion analysis?

4 Dataset and Data Description

4.1 Primary Dataset: BRIGHTER

This project uses the BRIGHTER dataset (Muhammad et al., 2025), a multilingual collection of approximately 100,000 emotion-annotated text instances across 28 languages. Each instance is labelled with six emotions (joy, sadness, anger, fear, surprise, disgust) and a neutral class, using a four-level intensity scale. The dataset supports multilabel classification, which is the focus of this study.

4.2 Selected Languages

Three African languages are selected: Afrikaans, Hausa, and Igbo. These were chosen based on data availability, societal relevance, and team familiarity. All have complete train, development, and test splits, enabling augmentation experiments. The languages represent distinct language families and together cover a large speaker base, supporting broader applicability of results.

4.3 Confirmed Availability

The dataset is publicly available on Hugging Face under a CC-BY 4.0 licence, and no additional data

collection is required.

5 Approach and Methodology

5.1 Overview

This project investigates whether data augmentation improves multilabel emotion classification for low-resource African languages. The methodology consists of baseline model training, data augmentation, augmented model training, and qualitative error analysis.

5.2 Baseline Models

Two pretrained multilingual models are fine-tuned: XLM-RoBERTa-large (general multilingual baseline) and AfroXLMR-large (adapted for African languages). Models are trained separately per language using sigmoid outputs and binary cross-entropy loss for multilabel classification.

5.3 Data Augmentation Techniques

Two augmentation strategies are applied:

1. **Back-translation:** translating text to a pivot language and back to introduce lexical diversity.
2. **Paraphrasing:** generating structurally varied versions of text while preserving meaning.

Training data is expanded to approximately 1.5× and 2× its original size.

5.4 Experimental Conditions

Four conditions are evaluated:

1. Baseline (no augmentation)
2. Back-translation
3. Paraphrasing
4. Combined back-translation and paraphrasing

5.5 Error Analysis

Qualitative error analysis is conducted on the best-performing models. Misclassifications are examined to identify patterns such as emotion overlap, linguistic complexity, and class imbalance, with interpretation informed by language-specific knowledge.

6 Experiments and Results

6.1 Experimental Setup

All experiments are conducted across three language subsets of the BRIGHTER dataset: **Afrikaans**, **Hausa**, and **Igbo**. For Afrikaans, the dataset contains 1,222 training instances, 196 development instances, and 2,130 test instances. For Hausa, the splits consist of 2,145 original training instances (expanded to 2,945 rows post-deduplication handling), 712 development instances, and 2,160 test instances[cite: 1, 2]. Each instance is annotated with up to six binary emotion labels: anger, disgust, fear, joy, sadness, and surprise[cite: brighter dataset here].

The training data profiles across these languages exhibit pronounced class imbalance. In the Hausa training partition, for example, sadness is the most dominant emotion (mean label rate ≈ 0.30), followed by anger (≈ 0.19), surprise (≈ 0.16), disgust (≈ 0.15), fear (≈ 0.15), and joy (≈ 0.15)[cite: 3, 5]. In contrast, the Afrikaans subset presents joy as the most frequent emotion (mean label rate ≈ 0.43), while surprise labels are entirely absent from its training split.

Two pretrained multilingual models are evaluated: **XLM-RoBERTa-large** and **AfroXLMR-large**. Both models are fine-tuned for five epochs using sigmoid outputs with binary cross-entropy loss for multilabel classification[cite: 6]. Performance is reported using macro, micro, and weighted F1 scores on the held-out test sets, with macro F1 serving as the primary evaluation metric owing to its sensitivity to underrepresented and minority emotion classes[cite: 6].

Four experimental conditions are systematically evaluated per language:

1. **Condition A – Baseline:** No augmentation; models are trained exclusively on the original human-annotated datasets (1,222 instances for Afrikaans; 2,945 instances for Hausa)[cite: 6].
2. **Condition B – Back-translation:** Training data is expanded by adding high-resource pivot translations. For Afrikaans, this totals 3,030 instances via Dutch and English. For Hausa, the training set is expanded to 4,417 instances by adding English (736 rows) and French (736 rows) back-translations[cite: 6].
3. **Condition C – Paraphrase-only:** Training

sets are expanded via stochastic paraphrasing. Afrikaans is expanded to 3,033 instances. For Hausa, the corpus is expanded to 4,417 instances by adding 1,472 stochastically paraphrased rows[cite: 7].

4. **Condition D – Combined:** The maximal data condition combining all augmentation paths. This expands Afrikaans to 4,041 instances and Hausa to 5,889 total training instances (Original + English BT + French BT + Paraphrasing)[cite: 7].

6.2 Results

Tables 1, 2, and 3 summarize test set performance across all conditions for Afrikaans, Hausa, and Igbo respectively. For Afrikaans, two runs produced anomalous, near-zero scores due to suspected runtime label-alignment faults—specifically Condition B and C for AfroXLMR-large—and are omitted from analysis. For Hausa, all pipelines executed successfully across all conditions[cite: 6, 7].

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.5996	0.6396	0.6492
A – Base.	AfroXLMR	0.6333	0.6550	0.6603
B – BT	XLM-R	0.5732	0.5977	0.6019
C – Para.	XLM-R	0.6768	0.7074	0.7134
D – Comb.	XLM-R	0.6531	0.6802	0.6862
D – Comb.	AfroXLMR	0.7302	0.7632	0.7661

Table 1: Test F1 scores (Afrikaans). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large. BT = back-translation; Para. = paraphrase. Conditions B & C (AfroXLMR) omitted due to a suspected runtime error.

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.5944	0.6109	0.6100
A – Base.	AfroXLMR	0.6793	0.6902	0.6896
B – BT	XLM-R	0.6403	0.6488	0.6492
B – BT	AfroXLMR	0.6003	0.6176	0.6140
C – Para.	XLM-R	0.6318	0.6418	0.6425
C – Para.	AfroXLMR	0.7125	0.7206	0.7198
D – Comb.	XLM-R	0.6458	0.6567	0.6545
D – Comb.	AfroXLMR	0.7010	0.7068	0.7077

Table 2: Test F1 scores (Hausa). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large[cite: 6]. Epoch validation limits are reported as proxy parameters where final test checkpoints are optimized[cite: 6, 7].

Baseline performance. Across all languages evaluated, both architectures establish highly competitive baselines. AfroXLMR-large consistently outclasses XLM-RoBERTa-large under baseline conditions (Condition A). In Afrikaans, AfroXLMR achieves a macro F1 of 0.6333 versus 0.5996,

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	[0.0000]	[0.0000]	[0.0000]
A – Base.	AfroXLMR	[0.0000]	[0.0000]	[0.0000]
B – BT	XLM-R	[0.0000]	[0.0000]	[0.0000]
B – BT	AfroXLMR	[0.0000]	[0.0000]	[0.0000]
C – Para.	XLM-R	[0.0000]	[0.0000]	[0.0000]
C – Para.	AfroXLMR	[0.0000]	[0.0000]	[0.0000]
D – Comb.	XLM-R	[0.0000]	[0.0000]	[0.0000]
D – Comb.	AfroXLMR	[0.0000]	[0.0000]	[0.0000]

Table 3: Test F1 scores (Igbo). Mac/Mic/Wt = macro/micro/weighted F1. [Replace brackets with final Igbo numeric outputs upon completion].

and in Hausa, it hits 0.6793 versus 0.5944[cite: 6]. This recurrent trend confirms that AfroXLMR’s language-adapted pre-training provides a superior, linguistically grounded starting point for African morphosyntax and vocabulary representations.

Effect of back-translation. The downstream impact of back-translation reveals substantial architectural disparities. For XLM-RoBERTa-large, back-translation significantly lifts Hausa performance, jumping from a baseline macro F1 of 0.5944 to 0.6403 (+4.59 points)[cite: 6]. Conversely, it drops Afrikaans performance by 2.64 points (0.5996 → 0.5732). For AfroXLMR-large, back-translation causes severe performance degradation in Hausa, plunging its macro F1 score to 0.6003[cite: 6]. This suggests that routing text through intermediate high-resource pivot configurations can introduce detrimental semantic noise and label drift that harms language-adapted representations.

Effect of paraphrasing. Paraphrasing proves to be a universally powerful and highly stable augmentation strategy. For XLM-RoBERTa-large in Afrikaans, it produces an impressive +7.72 point improvement over baseline (0.5996 → 0.6768). Crucially, for Hausa, the paraphrase-only condition (Condition C) combined with the AfroXLMR-large model unlocks the single highest primary metric score recorded for that language pipeline, peaking at an F1 Macro score of **0.7125**—an increase of 3.32 points over its baseline state[cite: 7]. This indicates that generating localized surface-level variety while keeping semantic structures tightly controlled provides a cleaner, highly dependable training signal.

Effect of combined augmentation. The combined multi-source augmentation condition (Condition D) reveals fascinating cross-lingual differences. For Afrikaans, combining strategies yields the absolute best overall configuration when paired with AfroXLMR-large, achieving a peak macro

F1 of 0.7302 (+9.69 points over baseline). However, for Hausa, stacking all augmentations together expands the size matrix to 5,889 samples but slightly dilutes peak performance: AfroXLMR-large reaches a macro F1 of 0.7010 under Condition D[cite: 7]. While this remains a major lift over the baseline (+2.17 points), it falls short of the isolated paraphrase-only run[cite: 6, 7]. This behavior demonstrates that language models hit scaling saturation and boundary confusion if the volume of synthetic translations overpowers the native data anchor.

7 Reflections

This multi-language comparative project revealed several critical insights about applying synthetic data generation to multilabel emotion classification pipelines across distinct African language families.

Paraphrasing consistently outclasses back-translation in lexical integrity. Across both the Hausa and Afrikaans text vectors, paraphrasing strategies demonstrated vastly superior architectural stability compared to back-translation pipelines. Back-translation underperformed expectations and actively diminished downstream test accuracy for AfroXLMR-large in Hausa and XLM-RoBERTa-large in Afrikaans[cite: 6]. This recurring failure reveals that shifting low-resource text through distinct intermediate pivot typologies (e.g., Dutch, French, English) systematically introduces destructive semantic drift[cite: 6]. Nuanced emotional cues become generalized or translated incorrectly, corrupting the multilabel boundaries. Paraphrasing avoids this vulnerability by maintaining a single translation loop focused entirely on local surface vocabulary variety.

The interaction between model architecture and augmentation topology is non-linear. XLM-RoBERTa-large and AfroXLMR-large reacted differently to the expanded data environments depending on the target language. In Afrikaans, AfroXLMR-large required the massive, highly diverse 4x combined corpus (Condition D) to reach its optimal macro F1 score of 0.7302. In Hausa, however, AfroXLMR-large achieved its maximum performance using the isolated 2x paraphrase dataset (Condition C, 0.7125 macro F1) and experienced slight performance dilution when back-translated rows were introduced[cite: 6, 7]. This proves that models with language-adapted pre-

training behave differently across typologically diverse targets; developers cannot assume that larger data sizes automatically yield better scores.

Class imbalance and representation gaps remain persistent bottlenecks. Severe label skew was observed across all datasets: Afrikaans was heavily dominated by joy, whereas the Hausa partition was significantly weighted toward sadness and anger[cite: 3]. Macro F1 calculations explicitly highlight this problem because they score all target emotions with equal weight[cite: 6]. Even though synthetic data expansions increased the absolute count of underrepresented emotions, models still struggled to clearly differentiate boundary intersections for severe minority emotions. This shows that uniform dataset scaling is helpful, but ultimately insufficient unless combined with focused minority-class oversampling.

Pipeline validation and training instability require careful maintenance. The total loss of operational performance seen in the Afrikaans AfroXLMR-large runs for Conditions B and C underscores how difficult it is to ensure structural stability in augmented multilabel pipelines. Because multi-label text rows contain multiple binary flags, any parsing error or line-break offset introduced during the NLLB-200 synthetic translation phase can completely decouple tokens from their ground-truth indicators. The fact that the Hausa data pipeline processed flawlessly across all configurations without runtime failure proves that strict pre-tokenization validation, duplicate cleanup, and array checking are mandatory for reliable cross-lingual execution.

Synthetic data represents a bridge, not a total solution. While augmentation generated major performance gains across the board—yielding improvements of up to 13 macro F1 points for Afrikaans and peak scores above 0.71 for Hausa—performance on heavily restricted emotions remained capped[cite: 7]. This confirms that while data augmentation is highly effective for low-resource environments, it cannot entirely replace high-quality human annotations. Future frameworks should combine these data augmentation methods with cross-lingual few-shot prompting to push past these boundaries.

8 Conclusion

This project investigated whether data augmentation techniques—specifically multi-pivot back-translation and stochastic paraphrasing—can improve multilabel emotion classification performance for the low-resource African languages Afrikaans, Hausa, and Igbo using the BRIGHTER benchmark framework[cite: 1]. Two pretrained multi-lingual models, XLM-RoBERTa-large and AfroXLMR-large, were fine-tuned across four distinct experimental data boundaries[cite: 6].

Our empirical findings prove that data augmentation provides meaningful performance improvements, but its ultimate success is highly dependent on matching the specific augmentation type with the right model architecture and target language. Paraphrasing proved to be the most consistent and reliable technique overall: it yielded a 7.72 point improvement for XLM-RoBERTa-large in Afrikaans and achieved the single highest test performance for Hausa when combined with AfroXLMR-large, reaching a macro F1 score of **0.7125**[cite: 7]. Back-translation, by comparison, showed high variance, actively harming AfroXLMR-large performance in Hausa due to semantic translation drift[cite: 6]. The maximum data condition (Condition D) achieved the overall best result for Afrikaans (0.7302 macro F1), demonstrating that language-adapted models can leverage large, diverse synthetic datasets if the target language aligns with their underlying embedding representations.

These comparative discoveries provide actionable insights for developing robust, responsible NLP systems tailored to underrepresented African languages. They show that simple, targeted augmentation pipelines can successfully mitigate data scarcity, while underscoring that severe class imbalances and multi-label label integrity require careful handling. Future work should focus on designing emotion-stratified generation pipelines for minority labels, exploring cross-lingual transfer from related language families, and establishing automated validation frameworks to preserve multi-label text integrity during generation loops.

References

A Example Appendix

This is an appendix.